

Experiment 03

Step-by-Step Equation Derivations

Op-Amp Integrator and Differentiator

This document explains and derives the main equations in Experiment 03 using a clear, visual, step-by-step approach.

0) Important Basic Op-Amp Ideas

Before deriving the equations, remember these two ideal op-amp rules:

Rule 1: Input current is almost zero

An ideal op-amp does not draw current into its input terminals.

$$I_+ \approx 0 \quad \text{and} \quad I_- \approx 0$$

Rule 2: Virtual Ground

In these circuits, the non-inverting terminal (+) is grounded ($V_+ = 0\text{V}$). Because of negative feedback, the inverting terminal (−) matches it:

$$V_- \approx V_+ \implies V_- \approx 0 \text{ V}$$

The inverting terminal behaves like ground, even though it is not physically connected to it. This is called a **virtual ground**.

1 Useful Component Equations

1.1 Resistor Current

For a resistor, $I = \frac{V}{R}$. If the input voltage is V_{in} and the inverting input is at virtual ground (0V):

$$I_R = \frac{V_{in} - 0}{R} \implies I_R = \frac{V_{in}}{R}$$

1.2 Capacitor Current

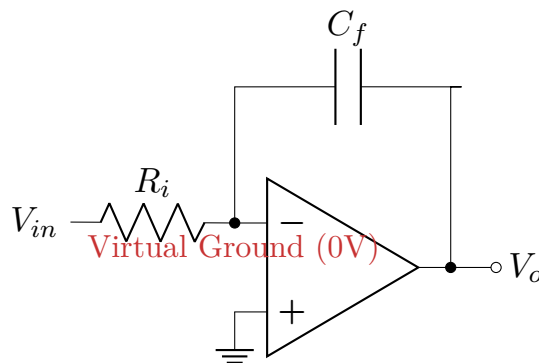
Capacitor current depends on how fast voltage changes:

$$I_C = C \frac{dV}{dt}$$

For AC sinusoidal analysis, the capacitor impedance is $Z_C = \frac{1}{j\omega C}$, where $j = \sqrt{-1}$ and $\omega = 2\pi f$.

2 Part A - Op-Amp Integrator

2.1 Ideal Inverting Integrator Circuit



2.2 Deriving the Ideal Integrator Equation

Step 1: Current through input resistor

Because $V_- \approx 0$, the current is $I_i = \frac{V_{in}}{R_i}$.

Step 2: Same current flows through feedback capacitor

Since the op-amp draws no current, I_i must flow through C_f . Thus, $I_i = I_C$.

Step 3: Capacitor voltage

The voltage across the capacitor is $V_C = V_- - V_o = 0 - V_o = -V_o$.

Step 4: Current through capacitor

$$I_C = C_f \frac{dV_C}{dt} \implies I_C = -C_f \frac{dV_o}{dt}.$$

Step 5: Equate currents

$$\frac{V_{in}}{R_i} = -C_f \frac{dV_o}{dt} \implies \frac{dV_o}{dt} = -\frac{V_{in}}{R_i C_f}.$$

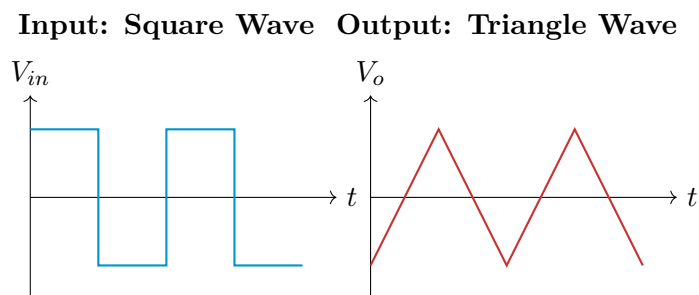
Step 6: Integrate both sides

Final Ideal Integrator Equation

$$V_o = -\frac{1}{R_i C_f} \int V_{in} dt$$

Meaning: The output is proportional to the **area under the input waveform**. The minus sign indicates inversion.

2.3 Waveform Transformations



- **Sine Input** $\rightarrow$ Output is a Cosine wave (amplitude $\propto 1/f$).

2.4 Practical Integrator

A feedback resistor R_f is added in parallel with C_f to prevent DC saturation.

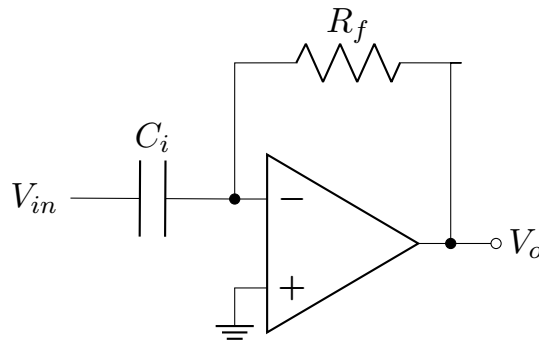
Practical Integrator Transfer Equation

$$\frac{V_o}{V_{in}} = -\frac{R_f}{R_i} \left[\frac{1}{1 + j\omega R_f C_f} \right]$$

Proper integration requires $\omega R_f C_f \gg 1$, meaning $f \gg \frac{1}{2\pi R_f C_f}$.

3 Part B - Op-Amp Differentiator

3.1 Ideal Inverting Differentiator Circuit



3.2 Deriving the Ideal Differentiator Equation

Step 1: Input capacitor current

Since $V_- \approx 0$, $V_C = V_{in}$. The current is $I_C = C_i \frac{dV_{in}}{dt}$.

Step 2: Same current flows through feedback resistor

$$I_f = \frac{0 - V_o}{R_f} = -\frac{V_o}{R_f}.$$

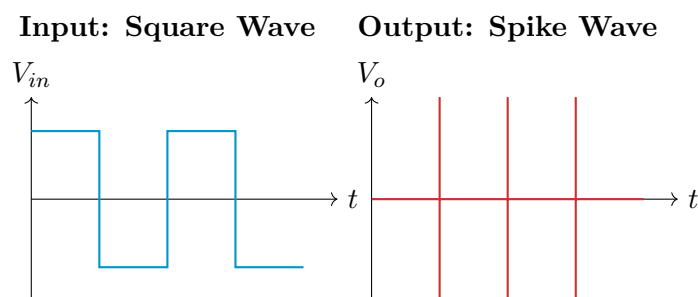
Step 3: Equate currents

Final Ideal Differentiator Equation

$$V_o = -R_f C_i \frac{dV_{in}}{dt}$$

Meaning: The output depends on the **rate of change** of the input. Fast change = large output.

3.3 Waveform Transformations



- **Triangle Input** → Output is a Square wave.
- **Sine Input** → Output is a Cosine wave (amplitude $\propto f$).

3.4 Practical Differentiator

A small resistor R_i is added in series with C_i , and a capacitor C_f is placed in parallel with R_f to limit high-frequency noise.

Practical Differentiator Transfer Equation

$$V_o = - \left[\frac{j\omega C_i R_f}{(1 + j\omega C_i R_i)(1 + j\omega C_f R_f)} \right] V_{in}$$

Proper differentiation occurs in the middle frequency range: $\frac{1}{2\pi R_f C_i} \ll f \ll \frac{1}{2\pi R_i C_f}$.

4 Summary Cheat Sheet

Circuit	Main Equation	Meaning & Frequency Effect
Integrator	$V_o = -\frac{1}{R_i C_f} \int V_{in} dt$	Output depends on <i>area</i> . Higher $f \rightarrow$ smaller output.
Differentiator	$V_o = -R_f C_i \frac{dV_{in}}{dt}$	Output depends on <i>rate of change</i> . Higher $f \rightarrow$ larger output.